

Hadoop Installation (Single Node Setup)

Step 1: Update Ubuntu

```
sudo apt update  
sudo apt upgrade -y
```

Step 2: Install Java

Check available Java version:

```
java -version
```

Install OpenJDK:

```
sudo apt install openjdk-11-jdk -y
```

Verify installation:

```
java -version
```

Find Java path:

```
readlink -f $(which java)
```

Example output:

```
/usr/lib/jvm/java-11-openjdk-amd64/bin/java
```

JAVA_HOME will be:

```
/usr/lib/jvm/java-11-openjdk-amd64
```

Step 3: Create Hadoop User

```
sudo adduser hadoop
```

Give password and press Enter.

Add user to sudo group:

```
sudo usermod -aG sudo hadoop
```

Switch to Hadoop user:

```
su - hadoop
```

Step 4: Configure SSH

Install SSH:

```
sudo apt install openssh-server openssh-client -y
```

Generate SSH key:

```
ssh-keygen -t rsa -P ""
```

Press Enter for all prompts.

Enable passwordless SSH:

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
chmod 600 ~/.ssh/authorized_keys
```

Test SSH:

```
ssh localhost
```

Type:

```
yes
```

Exit:

```
exit
```

Step 5: Download Hadoop

Move to home directory:

```
cd ~
```

Download Hadoop:

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz
```

Extract Hadoop:

```
tar -xvzf hadoop-3.4.1.tar.gz
```

Rename folder:

```
mv hadoop-3.4.1 hadoop
```

Step 6: Configure Environment Variables

Open bashrc:

```
nano ~/.bashrc
```

Add these lines at the end:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_HOME=$HOME/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
```

Save and exit:

Ctrl + O

Enter

Ctrl + X

Apply changes:

```
source ~/.bashrc
```

Step 7: Configure hadoop-env.sh

Open file:

```
nano ~/hadoop/etc/hadoop/hadoop-env.sh
```

Find:

```
export JAVA_HOME=
```

Replace with:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Save and exit.

Step 8: Configure core-site.xml

Open file:

```
nano ~/hadoop/etc/hadoop/core-site.xml
```

Replace content between `<configuration>` tags:

```
<configuration>
```

```
<property>
```

```
<name>fs.defaultFS</name>
```

```
<value>hdfs://localhost:9000</value>
```

```
</property>
```

```
</configuration>
```

Save and exit.

Step 9: Configure hdfs-site.xml

Open file:

```
nano ~/hadoop/etc/hadoop/hdfs-site.xml
```

Replace content:

```
<configuration>
```

```
<property>
<name>dfs.replication</name>
<value>1</value>
</property>

<property>
<name>dfs.name.dir</name>
<value>file:///home/hadoop/hadoopdata/hdfs/namenode</value>
</property>

<property>
<name>dfs.data.dir</name>
<value>file:///home/hadoop/hadoopdata/hdfs/datanode</value>
</property>

</configuration>
```

Save and exit.

Step 10: Create HDFS Directories

```
mkdir -p ~/hadoopdata/hdfs/namenode
mkdir -p ~/hadoopdata/hdfs/datanode
```

Step 11: Configure mapred-site.xml

Copy template:

```
cp ~/hadoop/etc/hadoop/mapred-site.xml.template ~/hadoop/etc/hadoop/mapred-site.xml
```

Open file:

```
nano ~/hadoop/etc/hadoop/mapred-site.xml
```

Add:

```
<configuration>

<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
```

```
</property>
```

```
</configuration>
```

Save and exit.

Step 12: Configure yarn-site.xml

Open file:

```
nano ~/hadoop/etc/hadoop/yarn-site.xml
```

Add:

```
<configuration>
```

```
<property>
```

```
<name>yarn.nodemanager.aux-services</name>
```

```
<value>mapreduce_shuffle</value>
```

```
</property>
```

```
</configuration>
```

Save and exit.

Step 13: Format NameNode

```
hdfsnamenode -format
```

Successful output:

Storage directory has been successfully formatted

Step 14: Start Hadoop Services

Start HDFS:

```
start-dfs.sh
```

Start YARN:
start-yarn.sh

Step 15: Verify Running Services

jps

Step 16: Check Hadoop Version

hadoop version

Step 17: Access Hadoop Web Interfaces

NameNode UI:

http://localhost:9870

ResourceManager UI:

http://localhost:8088

Step 18: Stop Hadoop

Stop YARN:

stop-yarn.sh

Stop HDFS:

stop-dfs.sh

Quick Verification Commands

java -version

```
This message is shown once a day. To disable it please create the
/home/sky/.hushlogin file.
sky@MS97377:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-22.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-22.04.2-Ubuntu, mixed mode, sharing)
```

hadoop version

```
sky@MS97377:~$ hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
```

jps

```
sky@MS97377:~$ jps
1045 NameNode
1846 Jps
1592 ResourceManager
1704 NodeManager
1165 DataNode
1325 SecondaryNameNode
```

hdfsdfsadmin -report

```
sky@MS97377:~$ hdfs dfsadmin -report
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1016675553280 (946.85 GB)
DFS Remaining: 1016670801920 (946.85 GB)
DFS Used: 4751360 (4.53 MB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
-----
Live datanodes (1):
Name: 127.0.0.1:9866 (localhost)
Hostname: MS97377.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 4751360 (4.53 MB)
Non DFS Used: 94333000992 (8.79 GB)
DFS Remaining: 1016670769152 (946.85 GB)
DFS Used%: 0.00%
DFS Remaining%: 94.04%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Jul 26 23:09:59 IST 2026
Last Block Report: Sun Jul 26 23:08:44 IST 2026
Num of Blocks: 29
```

hdfsdfs -ls /

```
sky@MS97377:~$ hdfs dfs -ls /
Found 2 items
drwxr-xr-x - sky supergroup 0 2026-07-25 16:40 /tmp
drwxr-xr-x - sky supergroup 0 2026-07-25 16:37 /user
```